ACCOUNTING RECEIVABLE SYSTEM

UNIVERSITY OF MINDANAO
College of Engineering
Computer Engineering Program

In Partial Fulfillment of the Requirement for the course
CpE 211/L- Data Structures and Algorithm

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SOFTWARE DESCRIPTION

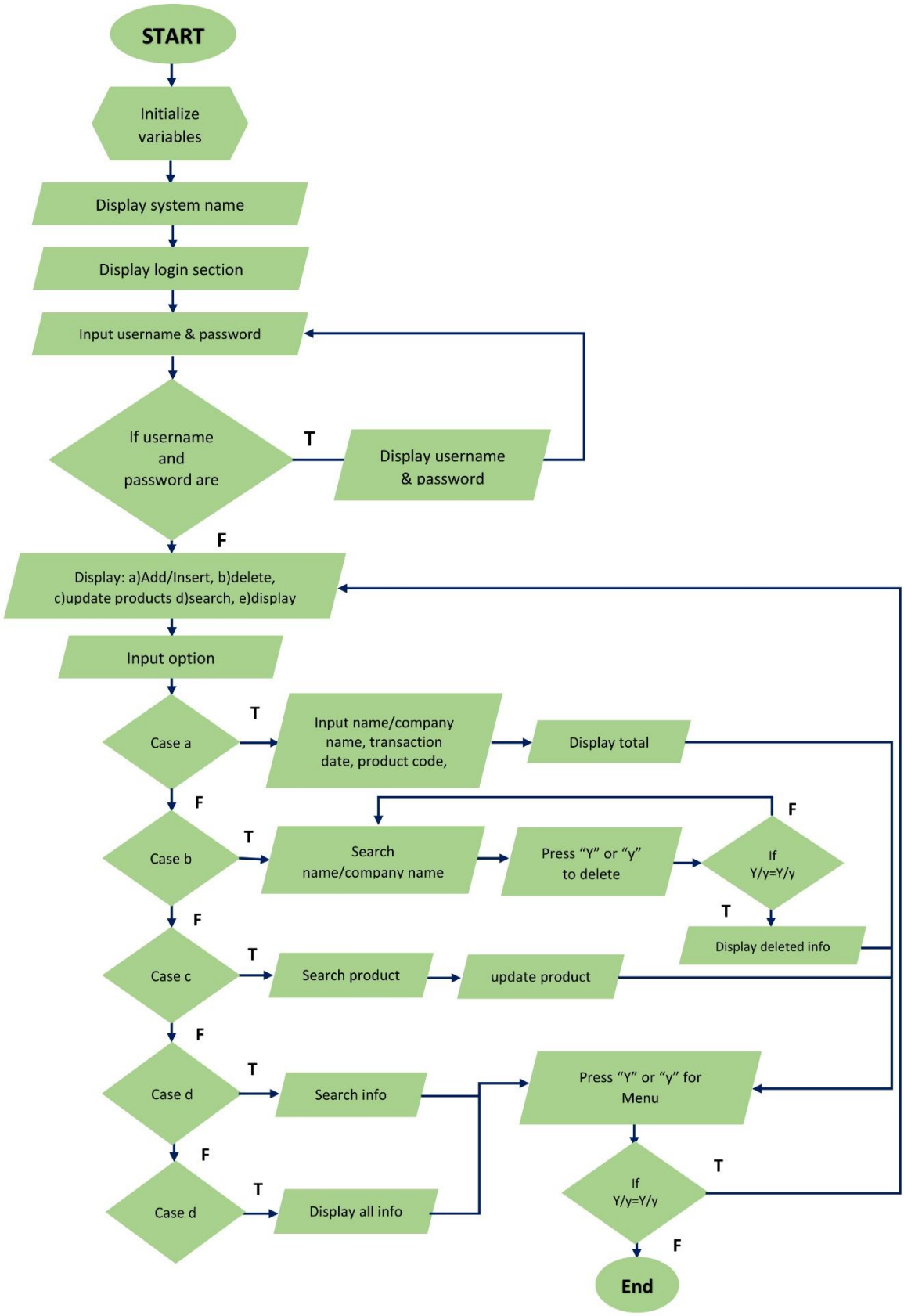
The Accounting Receivable System encompasses the recording of a customer's information and its purchased products. The system contains a menu with five buttons namely – Add, Products, Update, Search and Display. The Add Button enables the user to input its necessary details such as name, address, phone number and email. The customer can view all products available to purchase through the Products Button. It also views all the information of each products with its code, name, quantity, and price. The Update Button allows the user to update the customer's product and payment information. In searching some details, the user can proceed to the Search Button. Then, the Display Button views all stored details the customer had input as well as the payments. All details that were sent for input into the system shall be kept confidentially and secured.

Accounting Receivable System can keep track of details sent, payments received, and any customers that may be past due on their bills based on the indicated date. The customer can also customize its preferences based on their input to the Add Button. Though there can be limitations on the system. There are a few examples like - On the Search Button, you can only search via date. While on the Update Button, you can search via name or code.

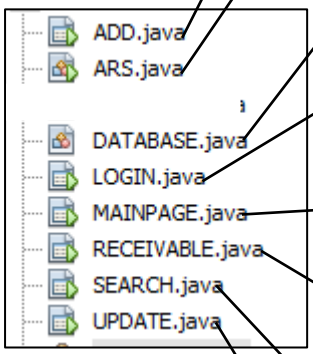
The system is surely created for comfort and with easy access among customers. It is user-friendly, with a clean design yet robust set of features that helps users generate actionable, data-based insights, and includes invoicing capabilities and receipt collection for good measure. Also, it is systematically fast, comprehensive, detailed for financial reports, and automated record keeping.

The platform requirements for the system, Accounting Receivable System are as follows: The OS should be Windows 7 with SP1; [Recommended: Windows 10]. The CPU must be Intel or AMD processor with 64-bit support; [Recommended: 2.8 GHz or faster processor]. The GPU must be nVidia GeForce GTX 1050 or equivalent; [Recommended: nVidia GeForce GTX 1660 or Quadro T1000]. The Disk Storage should have 4 GB of free disk space with a Monitor Resolution of 1280x800; [Recommended: 1920x1080].

FLOWCHART


START
Initialize variables
Display system name
Display login section
Input username & password
If username and password are
T
Display username & password
F
Display: a)Add/Insert, b)delete, c)update products d)search, e)display
Input option
Case a
T
Input name/company name, transaction date, product code,
Display total
F
Case b
T
Search name/company name
Press "Y" or "y" to delete
If Y/y=Y/y
F
T
Display deleted info
F
Case c
T
Search product
update product
F
Case d
T
Search info
Press "Y" or "y" for Menu
F
Case d
T
Display all info
If Y/y=Y/y
T
F
End

FILE DESCRIPTION



ADD Frame is where the customers information like name, address, etc. are being inputted.

ARS Class is the main class of the system where the LOGIN Frame can be called.

DATABASE Class is where the connection towards the database is being established.

LOGIN Frame is where the admin enters their id and password to access the system.

MAINPAGE Frame is the menu of the system where ADD, UPDATE, RECEIVABLE, SEARCH buttons are being displayed.

RECEIVABLE Frame is where we can view the monthly and yearly total receivables from the customers.

SEARCH Frame is where we can search and display customers information.

Update Frame is where we can update and delete customers information.

SOURCE CODE

```java
package ars;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import javax.swing.JOptionPane;
public class DATABASE {
    final static String JDBC_DRIVER = "com.mysql.jdbc.Driver";
    final static String DB_URL = "jdbc:mysql://localhost:3306/ars";

    final static String USER = "root";
    final static String PASS = "";
    public static Connection connection(){
        try{
            Class.forName(JDBC_DRIVER);
            Connection conn = DriverManager.getConnection(DB_URL,"root","");
            System.out.println("Connected");
            return conn;

        }catch(ClassNotFoundException | SQLException e){
            JOptionPane.showMessageDialog(null, e);
            return null;
        }

    }

}

```

DATABASE Code

```java

package ars;


public class ARS {

    public static void main(String[] args) {
        LOGIN gui =  new LOGIN();
        gui.setTitle("Login");
        gui.setLocationRelativeTo(null);
        gui.setVisible(true);
    }

}

```

ARS Class Code

```java
package ars;
import java.awt.HeadlessException;
import java.sql.Connection;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import javax.swing.JOptionPane;
public class LOGIN extends javax.swing.JFrame {
    Connection connect = null;
    Statement state = null;
    ResultSet result = null;
    public LOGIN() {
        super ("LOGIN");
        connect = DATABASE.connection();
        initComponents();
    }
    public void Login(){
        try {
            state = connect.createStatement();
            String ID = id.getText();
            String password = pass.getText();

            String sql = "SELECT * FROM login WHERE adminid = '"+ID+"'&& adminpass= '"+password+"' ";
            result = state.executeQuery(sql);
            if(result.next()){
                this.setVisible(false);
                String getter = ID;
                new MAINPAGE(getter).setVisible(true);
                }
            else{
                JOptionPane.showMessageDialog(null,"ID or PASSWORD is invalid.");
            }
        }
        catch (HeadlessException | SQLException e){
        }
    }
```

LOGIN Frame Code

```java
package ars;
import java.sql.Connection;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import javax.swing.JOptionPane;
public class MAINPAGE extends javax.swing.JFrame {
    Connection connect = null;
    Statement state = null;
    ResultSet result = null;
    String getter;
    public MAINPAGE( String ID) {
        initComponents();
        getter = ID;
        try{
            connect = DATABASE.connection();
            state = connect.createStatement();
            String sql = "SELECT * FROM login WHERE adminid = '"+ID+"'";
            result = state.executeQuery(sql);
            idl.setText(ID);
            while (result.next()){
                name.setText(result.getString(3).toUpperCase());
                if (result.getBoolean(3)){
                }
            }
        }catch(SQLException e){
            JOptionPane.showMessageDialog(null, e);
        }
    }

```

MAINPAGE Frame Code

```java
    private void jLabel9MouseClicked(java.awt.event.MouseEvent evt) {
        int input = JOptionPane.showConfirmDialog(null, "Confirm Logout?",null, JOptionPane.YES_NO_OPTION);
         if(input == JOptionPane.YES_OPTION) {
            this.setVisible(false);
            LOGIN gui = new LOGIN();
            gui.setTitle("Login");
            gui.setLocationRelativeTo(null);
            gui.setVisible(true);
         }
    }
```

LOGOUT Code

```java
    private void jLabel4MouseClicked(java.awt.event.MouseEvent evt) {

        new ADD().setVisible(true);
    }

    private void jLabel5MouseClicked(java.awt.event.MouseEvent evt) {

        new RECEIVABLE().setVisible(true);
    }

    private void jLabel4MouseEntered(java.awt.event.MouseEvent evt) {

    }

    private void jLabel6MouseClicked(java.awt.event.MouseEvent evt) {

        new SEARCH().setVisible(true);
    }

    private void jLabel2MouseClicked(java.awt.event.MouseEvent evt) {
        this.setVisible(false);
        new UPDATE().setVisible(true);
    }
```

JButton Function Call

```java
package ars;
import java.sql.Connection;
import java.sql.ResultSet;
import java.sql.PreparedStatement;
import java.sql.SQLException;
import java.text.SimpleDateFormat;
import javax.swing.JOptionPane;
public class ADD extends javax.swing.JFrame {
    Connection connect = null;
    PreparedStatement state = null;
    ResultSet result = null;
    String getter;
    public ADD() {
        initComponents();
    }
```

ADD Frame codes

```
203     private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
204         try{
205             connect = ars.DATABASE.connection();
206             String sql = "INSERT INTO `customerinfo`(`name`, `address`, "
207                     + "`phone`, `email`, `date`, `amount` ) VALUES (?,?,?,?,?,?)";
208             state = connect.prepareStatement(sql);
209             state.setString(1, cname.getText());
210             state.setString(2, caddress.getText());
211             state.setLong(3, Long.parseLong(cphone.getText()));
212             state.setString(4, cemail.getText());
213             SimpleDateFormat sdf = new SimpleDateFormat( "yyyy-MM-dd");
214             String date = sdf.format(pdate.getDate());
215             state.setString(5,date);
216             state.setFloat(6, Float.parseFloat(pamount.getText()));
217             state.executeUpdate();
218             JOptionPane.showMessageDialog(null, "Saved");
219             connect.close();
220 
221         }catch ( SQLException e){
222             JOptionPane.showMessageDialog(null, e);
223         }
224     }
```

Insert Code

```
226     private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
227         cname.setText(null);
228         caddress.setText(null);
229         cphone.setText(null);
230         cemail.setText(null);
231         pdate.setDate(null);
232         pamount.setText(null);
233     }
```

Clear Textbox Code

```
1   package ars;
2   import java.sql.Connection;
3   import java.sql.DriverManager;
4   import java.sql.PreparedStatement;
5   import java.sql.ResultSet;
6   import java.sql.ResultSetMetaData;
7   import java.sql.SQLException;
8   import java.sql.Statement;
9   import java.util.Vector;
10  import javax.swing.JOptionPane;
11  import javax.swing.table.DefaultTableModel;
12  import javax.swing.table.TableModel;
13  public class UPDATE extends javax.swing.JFrame {
14      Connection connect;
15      Statement state;
16      PreparedStatement pstate;
17      ResultSet result;
18      String getter;
19      public UPDATE() {
20          initComponents();
21      }
22         public Connection Connection(){
23         try{
24             connect = DriverManager.getConnection("jdbc:mysql://localhost/ars","root","");
25             return connect;
26         }
27         catch(SQLException e){
28             return null;
29         }
30     }
```

UPDATE Frame Code

```java
public final void ShowCustomers()
 {
     try{
         connect = ars.DATABASE.connection();
         state = connect.createStatement();
         String sql = "SELECT * FROM customerinfo ";
         result = state.executeQuery(sql);
         ResultSetMetaData rsm = result.getMetaData();
         int c = rsm.getColumnCount();
         DefaultTableModel df = (DefaultTableModel) this.utable.getModel();
         df.setRowCount(0);
         while (result.next()){
             Vector rowCell = new Vector();
             for (int i = 1; i < c; i++) {
                 rowCell.add(result.getString(1));
                 rowCell.add(result.getString(2));
                 rowCell.add(result.getString(3));
                 rowCell.add(result.getString(4));
                 rowCell.add(result.getString(5));
                 rowCell.add(result.getString(6));
                 rowCell.add(result.getString(7));
             }
             df.addRow(rowCell);
         }
         result.close();
     }catch(SQLException e){
         JOptionPane.showMessageDialog(null, e);
     }
 }
```

Show All Data in Table Code

```java
private void usearchActionPerformed(java.awt.event.ActionEvent evt) {
    String name = utext.getText().toUpperCase().replaceAll("\\s+", "");
    try{
        connect= ars.DATABASE.connection();
        state = connect.createStatement();
        String sql = "SELECT * FROM customerinfo WHERE id = '"+name+"' OR name LIKE '%"+name+"%'";
        result = state.executeQuery(sql);
        ResultSetMetaData rsm = result.getMetaData();
        int c = rsm.getColumnCount();
        DefaultTableModel df = (DefaultTableModel) this.utable.getModel();
        df.setRowCount(0);
        while (result.next()){
            Vector rowCell = new Vector();
            for (int i = 1; i < c; i++) {
                rowCell.add(result.getString(1));
                rowCell.add(result.getString(2));
                rowCell.add(result.getString(3));
                rowCell.add(result.getString(4));
                rowCell.add(result.getString(5));
                rowCell.add(result.getString(6));
                rowCell.add(result.getString(7));
            }
            df.addRow(rowCell);
        }
    }catch(SQLException e){
        JOptionPane.showMessageDialog(null, e);
    }
```

Search Data Code

```java
private void cdbuttonActionPerformed(java.awt.event.ActionEvent evt) {
    int row = utable.getSelectedRow();
    String a = utable.getModel().getValueAt(row, 0).toString();
    String sql = "DELETE FROM customerinfo WHERE id = " +a;
    try{
        pstate = connect.prepareStatement(sql);
        pstate.execute();
        JOptionPane.showMessageDialog(null, "Deleted Succesfully");
    }catch (Exception e){
        JOptionPane.showMessageDialog(null, e);
    }finally{
        try{
            pstate.close();
            result.close();
        }catch(Exception e){
        }
    }
    ShowCustomers();
    cname.setText(null);
    caddress.setText(null);
    cphone.setText(null);
    cemail.setText(null);
    cdate.setText(null);
    camount.setText(null);
    cid.setText(null);
}
```

Delete Code

```
388     private void cupbuttonMouseClicked(java.awt.event.MouseEvent evt) {
389         try{
390             connect = ars.DATABASE.connection();
391             String sql = "UPDATE customerinfo SET name= ?,address = ?,phone = ?,"
392                     + "email = ?,date = ?, amount = ? WHERE id = ?" ;
393             pstate = connect.prepareStatement(sql);
394             pstate.setString(7, cid.getText());
395             pstate.setString(1, cname.getText());
396             pstate.setString(2, caddress.getText());
397             pstate.setString(3, cphone.getText());
398             pstate.setString(4, cemail.getText());
399             pstate.setString(5, cdate.getText());
400             pstate.setString(6, camount.getText());
401             pstate.executeUpdate();
402             JOptionPane.showMessageDialog(null, "Updated Succesfully");
403             ShowCustomers();
404             }
405         catch (SQLException ex) {
406             JOptionPane.showMessageDialog(null, ex);
407             }
408             cname.setText(null);
409             caddress.setText(null);
410             cphone.setText(null);
411             cemail.setText(null);
412             cdate.setText(null);
413             camount.setText(null);
414             cid.setText(null);
415 
416     }
```

Update Data Code

```
418     private void utableMouseClicked(java.awt.event.MouseEvent evt) {
419         utable.setDefaultEditor(Object.class, null);
420         TableModel model = (TableModel)utable.getModel();
421         int a = utable.getSelectedRow();
422         cid.setText(model.getValueAt(a, 0).toString());
423         cname.setText(model.getValueAt(a, 1).toString());
424         caddress.setText(model.getValueAt(a, 2).toString());
425         cphone.setText(model.getValueAt(a, 3).toString());
426         cemail.setText(model.getValueAt(a, 4).toString());
427         cdate.setText(model.getValueAt(a, 5).toString());
428         camount.setText(model.getValueAt(a, 6).toString());
429     }
```

JTable Code

```
1   package ars;
2   import java.sql.Connection;
3   import java.sql.PreparedStatement;
4   import java.sql.ResultSet;
5   import java.sql.ResultSetMetaData;
6   import java.sql.SQLException;
7   import java.sql.Statement;
    import java.util.Vector;
9   import javax.swing.JOptionPane;
10  import javax.swing.table.DefaultTableModel;
11  import net.proteanit.sql.DbUtils;
12  public class RECEIVABLE extends javax.swing.JFrame {
13      Connection connect = null;
14      Statement state = null;
15      PreparedStatement pstate = null;
16      ResultSet result = null;
17      String getter;
18      public RECEIVABLE() {
19          initComponents();
20      }
```

RECEIVABLE Frame Code

```java
public void TotalAmount(){
    float sum = 0;
    for(float i =0; i<rtable.getRowCount(); i++){
        sum = sum + Float.parseFloat(rtable.getValueAt((int) i, 6).toString());
    }
    total.setText(Float.toString((float) sum));
}
```

Calculation of Total Amount Code

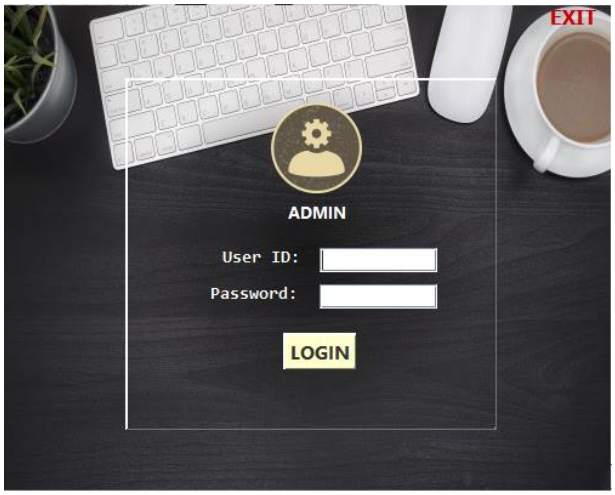
```java
package ars;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;
import java.sql.SQLException;
import java.sql.Statement;
import java.util.Vector;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;
import net.proteanit.sql.DbUtils;
public class SEARCH extends javax.swing.JFrame {
    Connection connect = null;
    Statement state = null;
    PreparedStatement pstate = null;
    ResultSet result = null;
    String getter;
    public SEARCH() {
        initComponents();
    }
public final void ShowCustomers()
    {
        try{
            connect = DATABASE.connection();
            String sql = "SELECT * FROM customerinfo";
            pstate = connect.prepareStatement(sql);
            result = pstate.executeQuery();
            stable.setModel(DbUtils.resultSetToTableModel(result));
        }catch(SQLException ex){
            JOptionPane.showMessageDialog(null, ex);
        }
    }
```

SEARCH Frame Code

```java
private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
    String name = search.getText().toUpperCase().replaceAll("\\s+", "");
     try{
         connect= ars.DATABASE.connection();
         state = connect.createStatement();
         String sql = "SELECT * FROM customerinfo WHERE id = '"+name+"' OR name LIKE '%"+name+"%'";
         result = state.executeQuery(sql);
         ResultSetMetaData rsm = result.getMetaData();
         int c = rsm.getColumnCount();
         DefaultTableModel df = (DefaultTableModel) this.stable.getModel();
         df.setRowCount(0);
         while (result.next()){
             Vector rowCell = new Vector();
             for (int i = 1; i < c; i++) {
                 rowCell.add(result.getString(1));
                 rowCell.add(result.getString(2));
                 rowCell.add(result.getString(3));
                 rowCell.add(result.getString(4));
                 rowCell.add(result.getString(5));
                 rowCell.add(result.getString(6));
                 rowCell.add(result.getString(7));
             }
             df.addRow(rowCell);
         }
     }catch(SQLException e){
         JOptionPane.showMessageDialog(null, e);
     }
 }
```

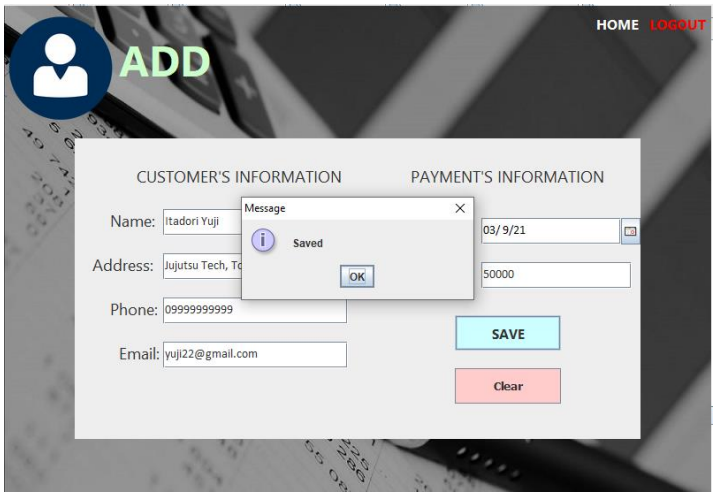
Search Code

SCREEN OUTPUT

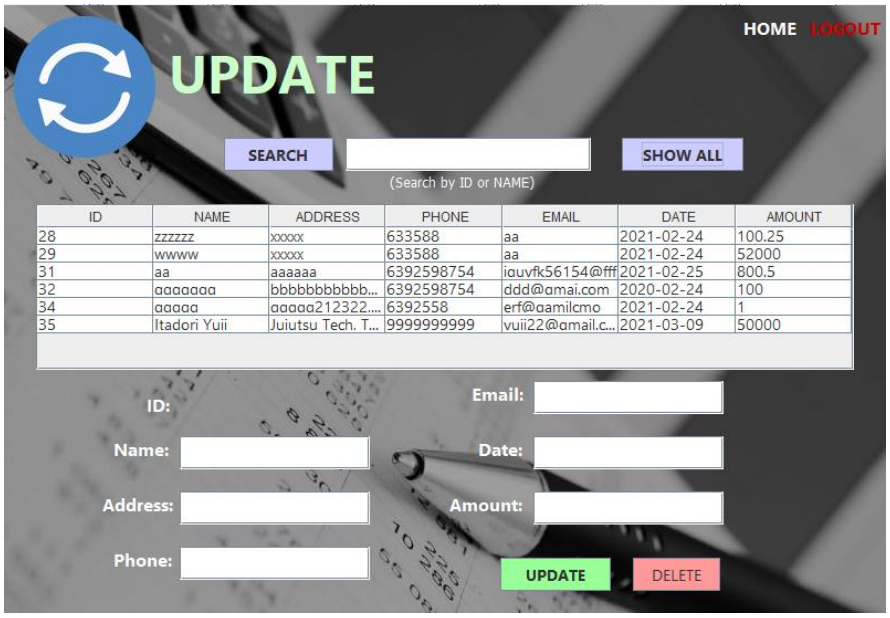


STEP 1

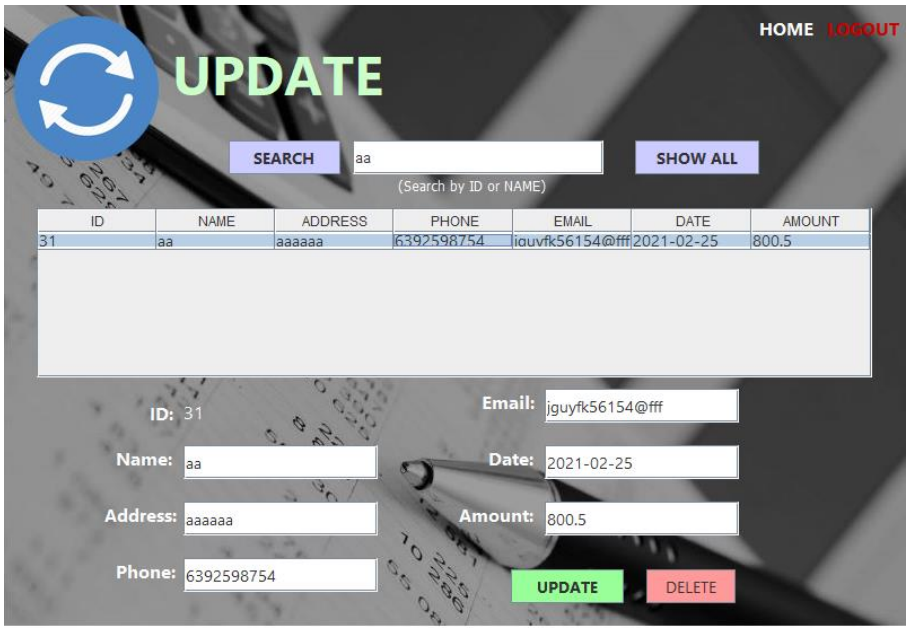
STEP 2



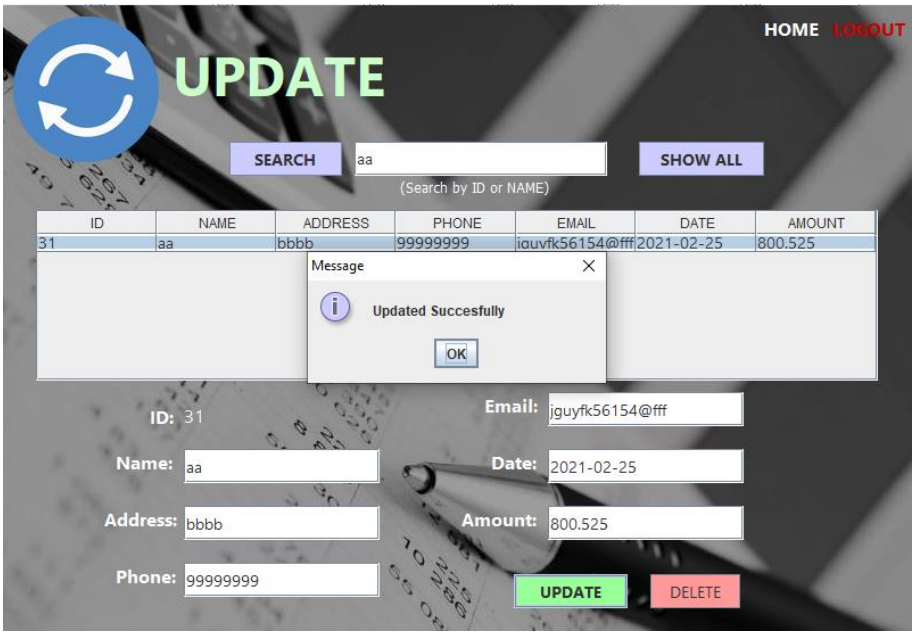
STEP 3



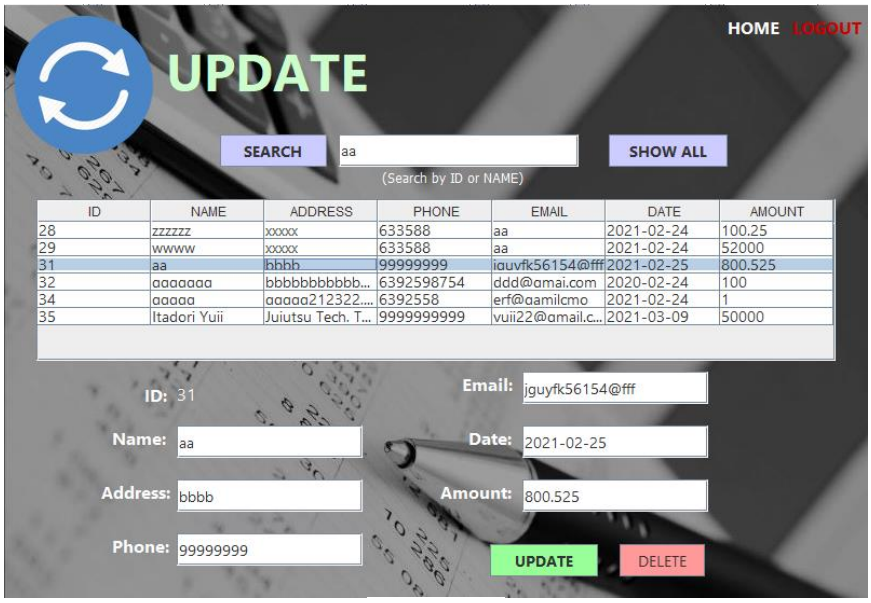
STEP 4



STEP 5



STEP 6



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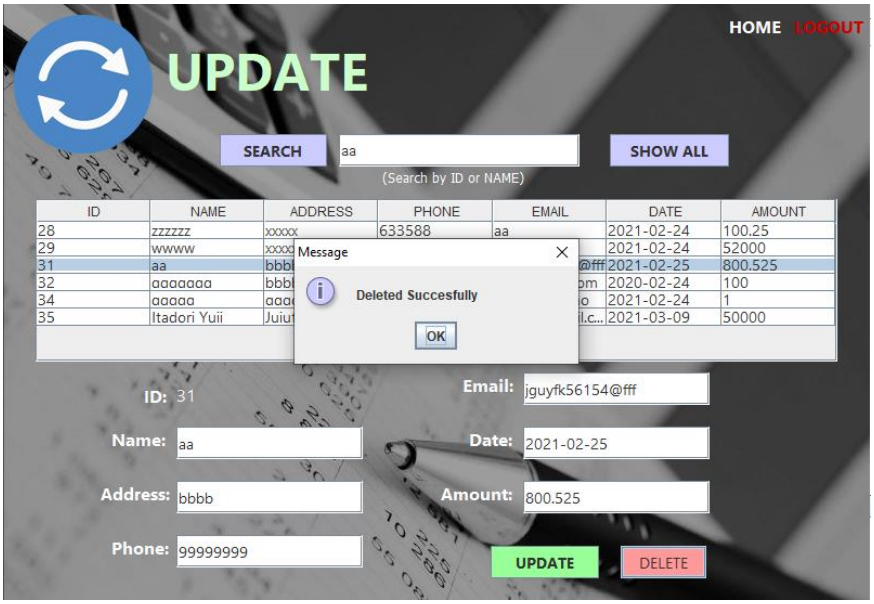
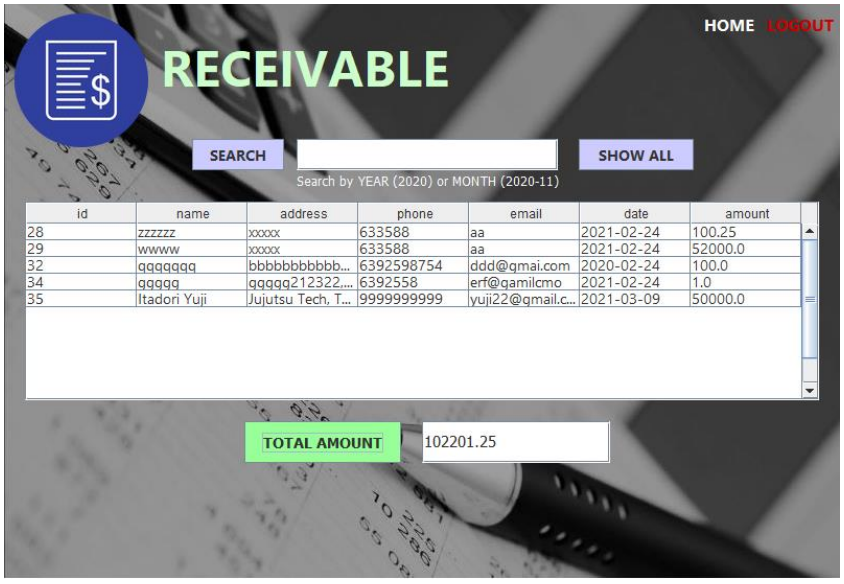


STEP 8



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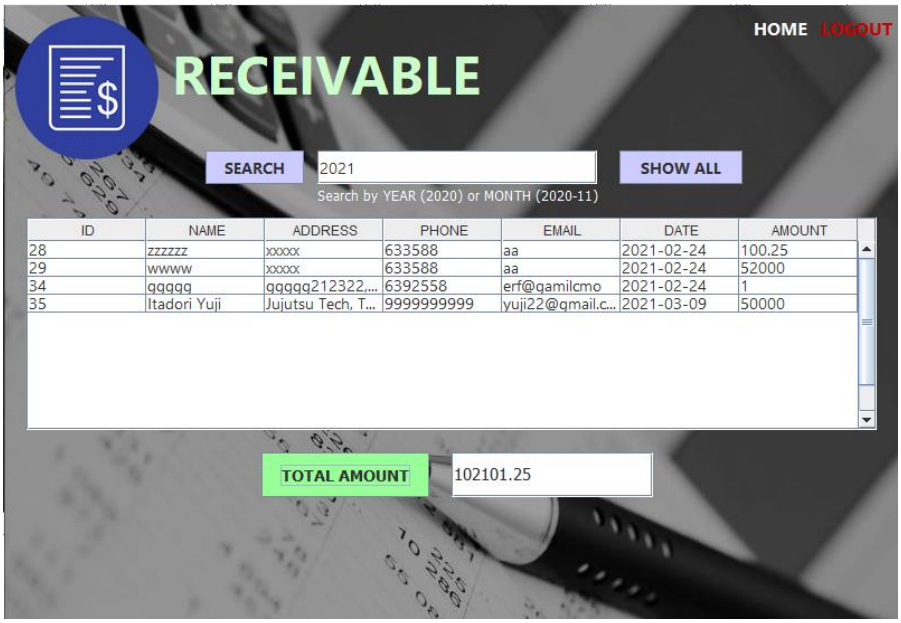

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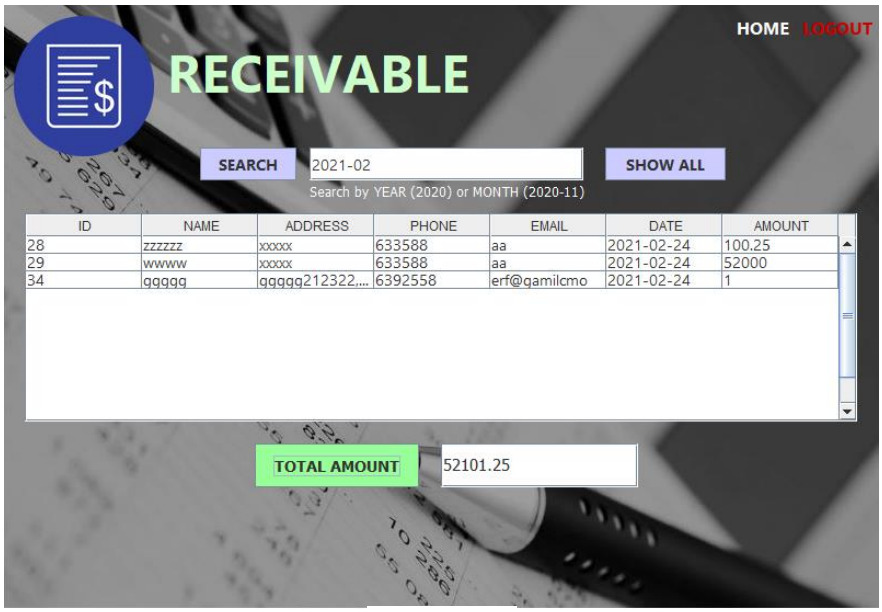

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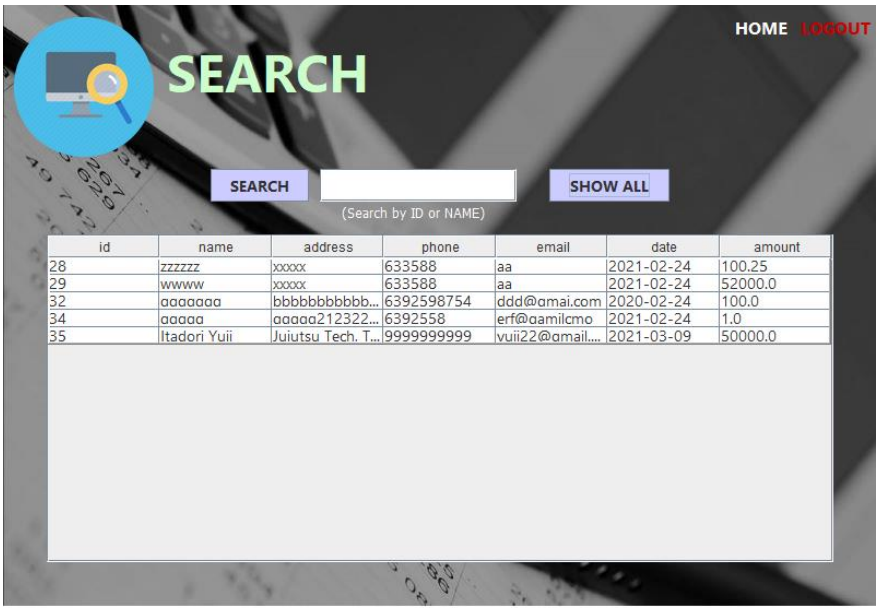
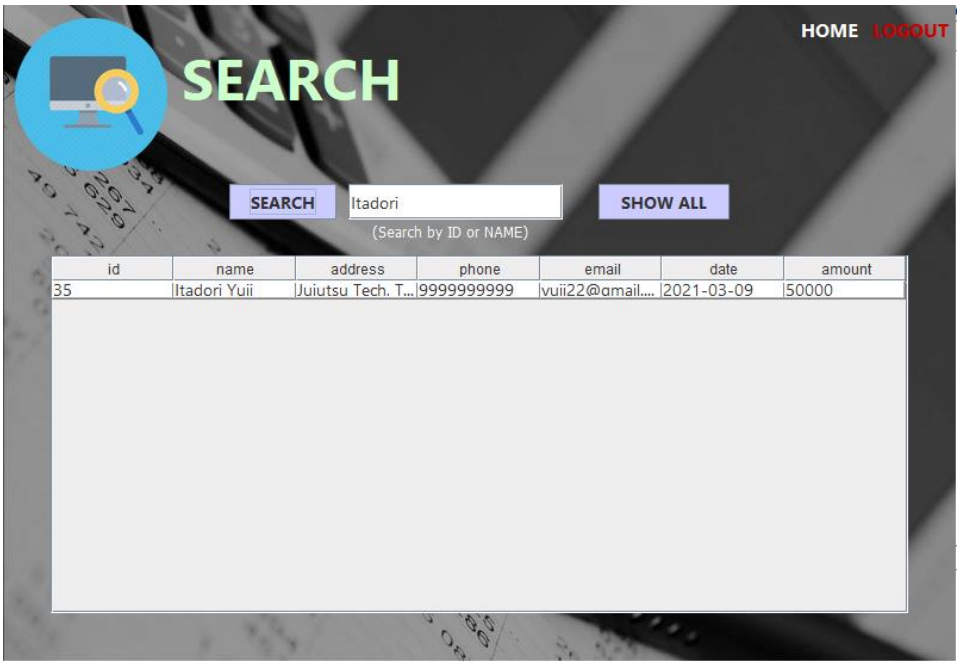
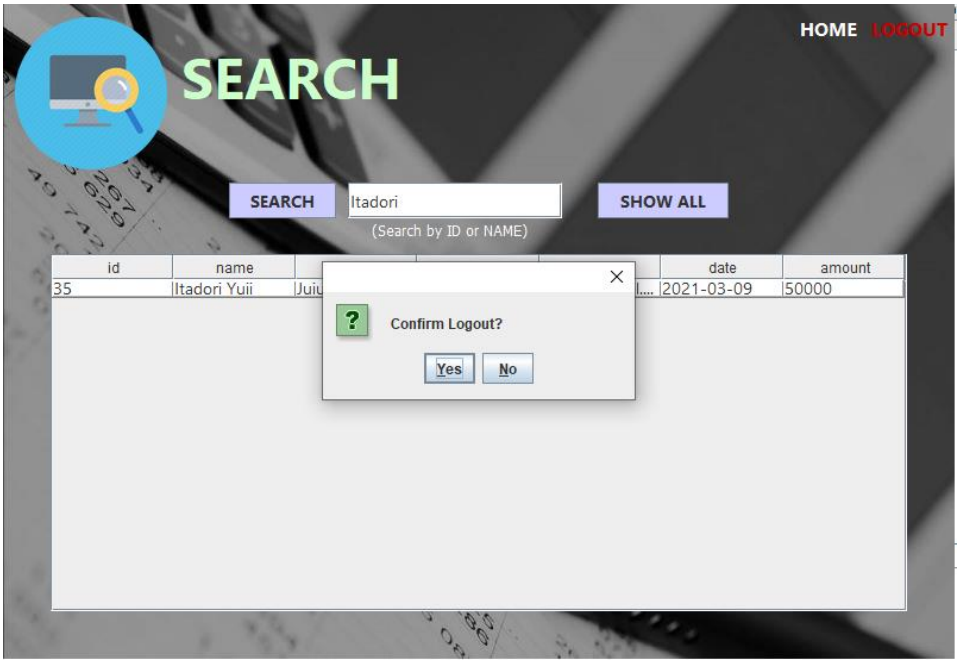


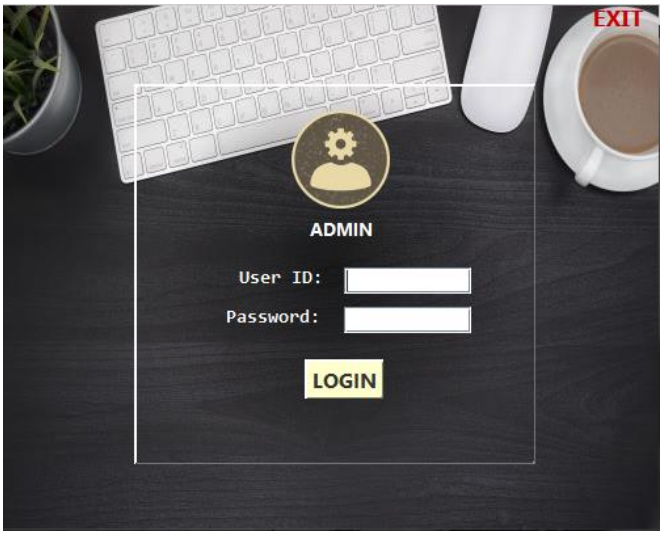
STEP 12



STEP 13



STEP 14



STEP 15

REFERENCES

https://www.researchgate.net/publication/322942500_Development_of_Account_Receivable_and_Payable_System_for_Travel_Bureau_Company

https://sba.thehartford.com/finance/accounting/accounts-receivable-process/

https://ofm.wa.gov/it-systems/accounts-receivable-system-ar